Start
Initialize NSGA-II settings and binary population
Apply decoder block masks
Evaluate population on proxy dataset
Generate captions
Compute objectives: Proxy CIDEr and pruning ratio
NSGA-II Update (Sorting, Crowding, Selection, Crossover, Mutation)
Create next generation
Proxy CIDEr improved?
YES
Update Proxy CIDEr
Reset patience counter
NO
Increase patience counter
Stopping criterion met? (max gen=50 or patience=20)
NO
Repeat next generation
YES
Extract final pareto front
Select LOW/MID/HIGH
Evaluation COCO Karpathy Test + NoCaps Val
End